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Palaeomagnetic Results from Lower Devonian Rocks of the Cheviot Hills, Northern England

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Summary

Palaeomagnetic work has been undertaken on four different rock types from the Cheviot Hills, Northern England. A total of 64 independently oriented handsamples from 21 different sites have been examined for their remanent magnetization. The rocks have been dated (K/Ar) as Lower Devonian. The most reliable palaeopole has been obtained from the combined results for the rhyolites and the andesites of the Cheviot volcanic series. It is situated at position 11° S, 140° E ($\alpha_{95} = 18°$, 11 sites, 35 handsamples, af cleaned). This pole and also some preliminary poles calculated for the metamorphosed andesites and the Cheviot Granite support the existence of a Lower Devonian group of palaeomagnetic north poles in the region of New Guinea as defined by previously published results from other British formations.

1. Introduction

There has been much discussion concerning the character of the geomagnetic field during the Devonian period, and about the position of the Devonian palaeomagnetic pole relative to Europe. Among many others, the papers by Creer (1964), Storetvedt & Halvorsen (1968), Embleton (1969, 1970), Creer (1970), and McElhinny and Briden (1971) have made it clear that some British rocks have come to occupy a central position in the problem, as they define a group of palaeomagnetic poles near the present day equator, whereas many other European rocks define a position different from that group of poles. This paper will give some additional data on the Devonian pole position relative to the British Isles, based on a palaeomagnetic investigation of some dated rocks from the Cheviot Hills.

2. Geology

The Cheviot Hills are gentle, peat covered mountains with a low degree of exposure. The igneous complex, which makes up the Cheviot Hills, has been described by Eastwood (1953) and Carruthers *et al.* (1932). The lowest bed of the volcanic series is an agglomerate, resting unconformably on upturned Silurian strata. Upon it follows flows of lavas of various compositions. The oldest flows are of rhyolitic character, whereas the younger ones are of andesitic composition. The latter make up the greater part of the massif. Due to the scarcity of exposures, single flows cannot easily be recognized, and the total thickness of the series is not known.

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Dykes and a large granite block were intruded into the lavas. The granite metamorphosed the surrounding lavas, producing pyroxene granulites and hornfelses (henceforth called metamorphosed andesites). At some places within the granite area small craggs of intensely metamorphosed andesites can be found. These are thought still to be *in situ* (Carruthers *et al.* 1932), and this suggests that elevated parts of the granite area represent the approximate position of the original roof of the intrusion, which does not seem to have resulted in any appreciable dooming or buckling of the lavas. The majority of the dykes predates the granite, but their mutual relationship is not clear. Some dykes follow crush zones (of which there are some in the area), and these may have been activated as late as the Tertiary (Dr Robson 1971, private communication). On the Scottish side of the border, the lavas are overlain by an Upper Devonian conglomerate containing boulders of both the Cheviot granite and the Cheviot lavas. The stratigraphic age of the Cheviot Hills can thus be inferred to be Lower Devonian (Carruthers *et al.* 1932).

Potassium–argon dates have been obtained from a variety of rocks from the Cheviot Hills (J. Mitchell 1972). Whole rock dating of the rhyolites, the andesites, and the metamorphosed andesites and biotite dating of the granite and some dykes yielded Lower Devonian ages, confirming the stratigraphic ages (Table 1). The granite therefore seems to belong to the suite of Newer Granites (Read & McGregor 1956) of late Caledonian age.

3. Experimental method

A total of 64 handsamples from 21 sites were collected. Magnetic compass and, whenever possible, Sun-compass bearings were taken, and these were generally in good agreement. Three of the sites were in the metamorphosed andesites, one was in the granite, and the rest in the rhyolitic and andesitic lavas. As most exposures were small, only two to five handsamples were taken at each site. Fig. 1 gives a geological sketch map with the location of sites used in the final pole calculations. Measurements of the magnetic remanence were performed on a P.A.R. SM2 spinner magnetometer. The stability of the remanence was tested by standard af demagnetization tests (e.g. Irving, Scott & Ward 1961). For a few sites the af treatment did not reduce the within-site scatter of directions, and such sites were not included in the final pole calculation. Only for one site (19) was it possible to apply a tilt correction, as only here the dip of beds could be determined with any certainty. However, the tilting of the lavabeds is nowhere drastic, it seldom exceeds 5°–8° and is mostly negligible (Carruthers *et al.* 1932). Furthermore, sampling was avoided in any of the crush zones. Curie temperatures were determined for one or two powdered samples from each site, using a translation balance (Creer & de Sa 1970).

Table 1

Summary of K/Ar-dating of rocks from the Cheviot Hills (Mitchell 1972)

Rock type	Analysed	Samples	Determinations	Age My*
Rhyolite	whole rock	1	2	378–383
Andesite	—	2	4	385–394
Granite	biotite	2	4	372–383
Metamorphosed andesites	whole rock	5	10	366–380
Dykes	biotite	2	4	370–388

* Ages are estimated accurate to between ±6 and ±9 My.

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FIG. 1. Geological sketch map of the Cheviot Hills, with location of sites.

4. Results

(a) *The andesitic and rhyolitic lavas*

Pilot demagnetization of the lavas was carried out in five steps up to 570 Oe peak on one specimen from each handsample (Fig. 2(a)–(d)). It was concluded that the lavas appeared to have a stable component of magnetization with a coercivity of remanence of at least 500–600 Oe peak. This stable component was in some cases masked by a softer component of magnetization with a coercivity of remanence in the range of 200–300 Oe peak. The scatter of directions within sites was at a minimum after cleaning at between 300 and 400 Oe peak. The bulk of the lava specimens was therefore demagnetized in a field of 350 Oe peak. This resulted in most cases in a clearly visible decrease of both within-site scatter (Fig.. 3) and between-site scatter (Fig. 4). In Table 2, the palaeomagnetic parameters of accepted sites after af demagnetization are given, and in Table 3, the palaeopole calculated from these sites is given. The NRM directions of site 19 were extremely stable, and no bulk demagnetization was performed on the samples from this site. The rhyolites exhibited two ranges of Curie temperatures, *viz.* 500–575 °C and 600–660 °C, indicating the presence of both titano-magnetites and haemoilmenites. The andesites showed Curie temperatures in the range 460–550 °C, and only in a few cases 600 – 650°C, indicating that titano-magnetites are the dominant magnetic mineral in this rock type.

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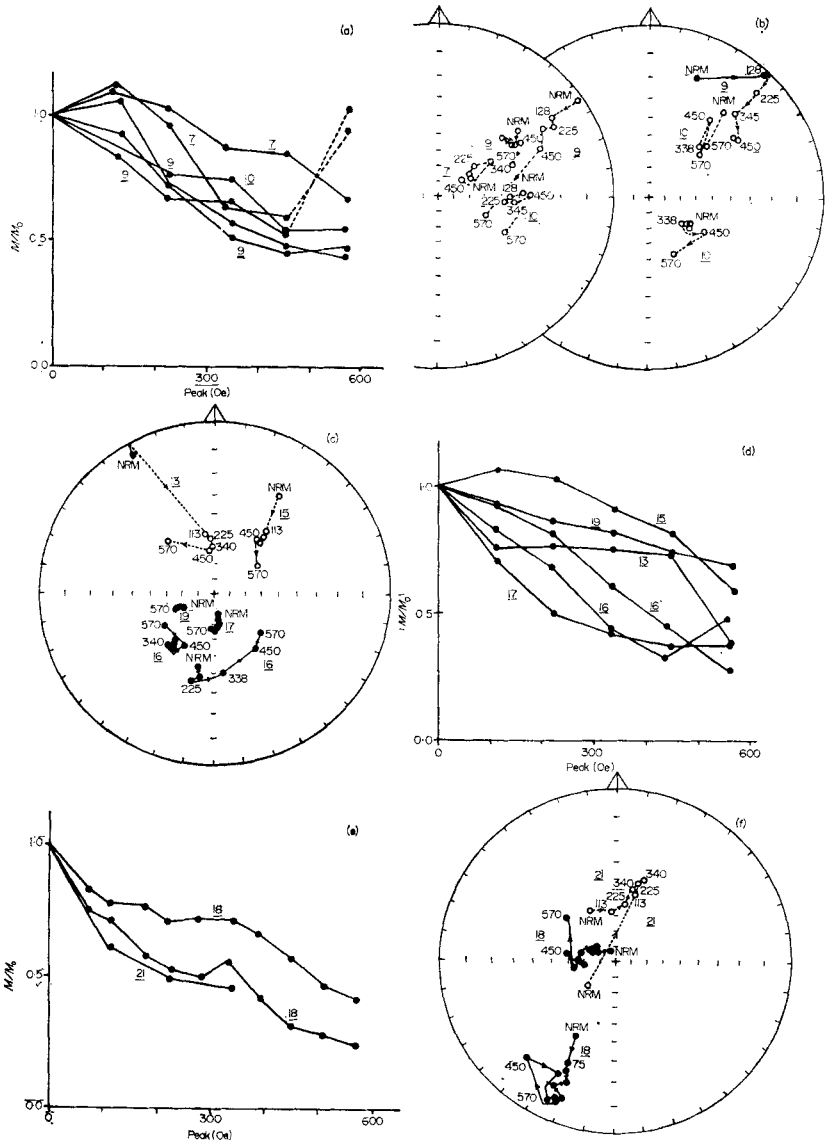


FIG. 2. AF demagnetization of typical samples of andesite (a–b), rhyolite (c–d), and granite and metamorphosed andesite (e–f). The underlined numbers give the number of the sampling site, the numbers in smaller print give the values of the demagnetizing peak fields. Closed symbols mean positive inclination; open symbols mean negative inclination.

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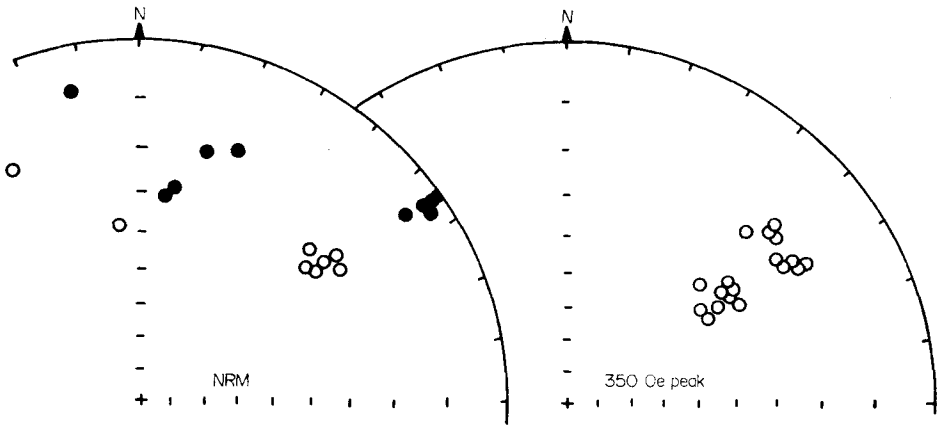


FIG. 3. Typical example of the reduced within-site scatter resulting from demagnetization at 350 Oe peak. The site is No. 9. Symbols as in Fig. 2.

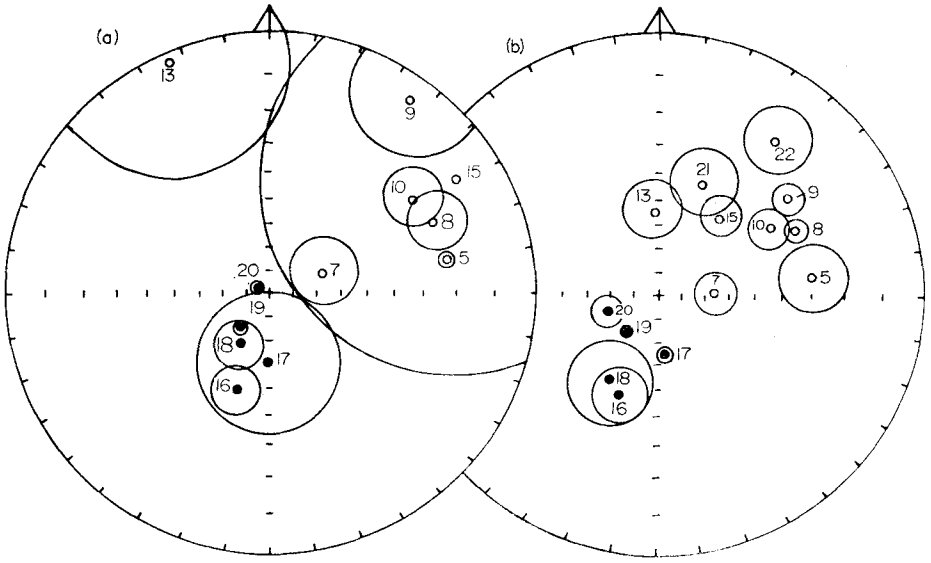


FIG. 4. Site means with circles of confidence, α_{95}, (a) before and (b) after demagnetization, clearly showing the decreased scatter. The NRM directions of sites 21 and 22 were almost randomly distributed and are not shown on the figure. Site 19 was not demagnetized (see text). Symbols as in Fig. 2.

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Table 2

Summary of af demagnetization

Site number	Rock type	Specimens	Peak field (Oe)	$D°$	$I°$	k	$\alpha_{95}°$
5	Andesite	5	350	84·2	−29·6	53·54	10·6
7	—	19	350	90·8	−65·3	18·63	8·0
8	—	11	350	65·9	−30·0	136·81	3·9
9	—	18	350	54·0	−27·3	57·29	4·6
10	—	12	350	60·6	−37·9	38·81	7·1
13	Rhyolite	6	350	357·0	−55·1	31·10	12·2
15	—	9	350	39·3	−48·3	58·85	6·8
16	—	7	350	198·8	45·5	28·56	11·5
17	—	16	350	171·6	63·6	148·67	3·0
19*	—	7	NRM	216·9	70·1	1759·71	1·4
20	—	11	350	250·8	68·3	57·33	6·1
(18	Granite	19	NRM	210·3	64·4	11·40	10·4)
18	—	19	125	207·0	48·8	6·21	16·7
(18	—	19	225	203·1	46·3	5·15	18·7)
21	Metamorphosed andesite	8	300	22·0	−41·6	23·92	11·6
22	—	8	300	36·3	−18·3	29·56	10·4

The standard palaeomagnetic notation for symbols is used (see e.g. Irving (1964).

* NRM stable.

The NRM intensity of the rhyolite ranged from 10^{-5} Gauss to 3×10^{-4} Gauss with a geometric mean of $1{\cdot}29\times10^{-4}$ Gauss. The andesites showed a wide range of NRM intensities, from 3×10^{-5} Gauss to 3×10^{-3} Gauss, the geometric mean being $1{\cdot}44\times10^{-4}$ Gauss. Table 3 gives the intensities of all the rock types after cleaning.

(*b*) *The metamorphosed andesites and the granite*

The NRM of the metamorphosed andesites exhibited a very soft component of viscous magnetization, which almost immediately aligned itself with any ambient field. As a consequence, the distribution of the NRM directions was practically random. With extreme care it was, however, possible to get an indication (af demagnetization, Fig. 2(e)–(f)) of an apparently stable component of magnetization, and after the subsequent bulk demagnetization carried out at 300 Oe peak, the directions from two sites grouped fairly well (Fig. 4). Tables 2 and 3 give the palaeomagnetic parameters. Thermomagnetic curves inferred Curie temperatures at 400 °C and 480 °C. The NRM intensity of the metamorphosed andesites was found to be quite high, ranging from 10^{-4} Gauss to 6×10^{-3} Gauss with a geometric mean of $9{\cdot}00\times10^{-4}$.

The result of pilot demagnetization of the granite specimens, done in ten steps up to 570 peak Oe (Fig. 2(e)–(f)), did not allow the choice of a single field intensity suitable for application to all the specimens, which were therefore demagnetized at 125 Oe peak and then at 225 Oe peak. After cleaning in both of these fields an increase in scatter of directions was found, but no change appeared in the general pattern of the distribution of directions as compared to the NRM directions, i.e. both groups were smeared along a great circle approximately through the present field direction. The palaeomagnetic parameters are given in Tables 2 and 3. The Curie temperatures of the granite (530 (?) and 625 °C) indicated the presence of both titano-magnetites and haemoilmenites. The NRM intensity was in the range 4×10^{-6} Gauss to 2×10^{-5} Gauss with geometric mean $8{\cdot}31\times10^{-6}$ Gauss.

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Table 3

Rock type	Number of sites	Number of hand samples	Mean direction of magnetization D	I	k	α_{95}	Range (log mean) of intensities after af cleaning (Gauss)	North Palaeopole position lat.	long.	α_{95}
Rhyolitic and Andesitic Lavas*	11	35	49·4	−53·0	10·41	14·9	$2\times10^{-6}-1\times10^{-3}$ $(7\cdot52\times10^{-5})$	11° S	140° E	18·0
Granite†	1	4	207·0	48·8	5·36	16·7	$1\times10^{-6}-1\times10^{-5}$ $(4\cdot53\times10^{-6})$	4° S	157° E	17·0
Metamorphosed Andesites‡	2	6	30·0	−30·1	16·05	9·5	$6\times10^{-5}-4\times10^{-4}$ $(2\cdot85\times10^{-4})$	13° N	150° E	7·8

* All lava sites (not metamorphosed andesite sites) of Table 1 included. Unit weight to sites.

† 125 Oe peak mean direction. Unit weight to specimens.

‡ The two metamorphosed andesite sites of Table 1. Unit weight to specimens.

It should be noted that in the author's opinion there are not sufficient data for the metamorphosed andesites nor for the Cheviot granite, and the palaeopoles calculated for these two formations should only be considered supplementary ones, not to be quoted as accepted Lower Devonian poles. Further details of the work on the rocks from the Cheviot Hills are given in Thorning (1971).

5. Discussion

Several British Devonian rock formations have been investigated for their palaeomagnetic properties (e.g. Creer 1970). Stubbs (1958) worked on Devonian lavas from the Midland Valley of Scotland and found a pole position (NRM) of 141° E, 10° S. Chamalaun & Creer (1964) obtained from thermally cleaned Anglo-Welsh cuvette samples a palaeopole situated at 117° E, 2° S. Embleton (1968) examined lavas from the Devonian rocks of the Lorne Plateau and from Tayside, Scotland. After af cleaning his measurements yielded a pole at 121° E, 1° N. McMurry (1970) carried out an extensive survey of Lower Devonian Scottish lavas using af cleaning and applying tilt corrections, and which gave a pole position of 145° E, 1° S. More recently Briden (1970) has reported pole positions from Caledonian intrusives and they lie at 144° E, 8° N, and 146° E, 5° N.

The work on Lower Devonian lavas from the Cheviot Hills reported in this paper yields a pole position of 140° E, 11° S. These poles are in substantial mutual agreement, and there seems no reason to doubt that during part of the Lower Devonian period, the palaeomagnetic pole relative to the British Isles was situated in the area suggested by the poles mentioned.

Storetvedt (1970, 1971) has argued that the direction obtained by Chamalaun & Creer (1964) for the Anglo-Welsh cuvette is not acceptable as a primary component of magnetization, and he also contends that the Scottish lavas studied by Stubbs (1958), and Embleton (1968), and McMurry (1970), have 'complex thermal histories' and consequently, that the directions of remanent magnetization obtained from them are not true primary directions of magnetization. Storetvedt & Halvorsen (1968) carried out thermal demagnetization experiments on some Scottish lavas (rhyolites?) from the Cheviot Hills in order to test the latter point of view. The direction of remanent magnetization of some of the samples they worked on showed that these had lost most of their primary magnetization and had been remagnetized in the direction of the present magnetic field, and the directions worked on cannot be said to be representative of the remanent magnetization of the rocks of the Cheviot massif. However, Storetvedt & Halvorsen (1968) contend that their results demonstrate that 'the Scottish lavas may be magnetically very complex and therefore unreliable from a paleomagnetic point of view', and Storetvedt (1970) maintains that the true Devonian pole position can best be recognized in certain rocks from Norway. The latter position is quite distinct from that established by the British rocks.

The concordance between the Devonian palaeomagnetic data now available from a variety of rock types collected over a wide area in Scotland and Northern England (including the present work on the Cheviot lavas) would, however, suggest that it is reasonable to conclude that the group of poles, to which the pole obtained in this work belongs, represents a real palaeomagnetic field over the areas sampled. This study of the lavas from the Cheviot Hills has not given any new indications as to over how broad an area this field existed, e.g. whether it might be observed in the Caledonian belt only.

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